SEQUENCES & SERIES

Arithmetic & Geometric Progressions

Grade 6 Mathematics

What Are Sequences & Series?

A **sequence** is a list of numbers arranged in a special order, following a pattern or rule.

A **series** is what we get when we add up the terms of a sequence.

Example:

2, 4, 6, 8, ... is a **sequence**

2 + 4 + 6 + 8 = 20 is a **series!**

What is an Arithmetic Progression (AP)?

Numbers **increase or decrease by the same amount** each time.

Formula:

$$a_n = a_1 + (n - 1)d$$

a_1	= first term
d	= common difference
n	= term number

Example of AP

Sequence: 3, 6, 9, 12, ...

Find the 5^{th} term:

$$a_5 = 3 + (5 - 1) \times 3$$

$$a_5 = 3 + 12 = 15$$

Sum of an AP Series

$$S_n = n/2 \times (2a_1 + (n - 1)d)$$

What is a Geometric Progression (GP)?

Numbers **multiply or divide by the same number** each time.

Formula:

$$a_n = a_1 \times r^{(n-1)}$$

a_1	= first term
r	= common ratio
n	= term number

Example of GP

Sequence: 2, 6, 18, 54, ...

Common ratio: r = 6 ÷ 2 = **3**

Sum of a GP Series

$$S_n = a_1 \times (r^n - 1) / (r - 1), \quad r \neq 1$$

Example: Sum of first 3 terms of 2, 6, 18

$$S_3 = 2 \times (3^3 - 1) / (3 - 1) = 2 \times 26/2 = 26$$

AP vs GP: What's the Difference?

Feature	AP	GP
Operation	Add / Subtract	Multiply / Divide
Growth type	Linear	Exponential
Key term	Common Difference (d)	Common Ratio (r)

n^{th} term	$a_1 + (n-1)d$	$a_1 \times r^{(n-1)}$
Example	2, 5, 8, 11, ...	2, 6, 18, 54, ...

Practice Problems: AP

1. Find the 7^{th} term of 5, 8, 11, ...
2. Find the sum of first 10 terms of 2, 4, 6, ...
3. Find the common difference in 15, 12, 9, ...

Practice Problems: GP

1. Find the 5^{th} term of 3, 9, 27, ...
2. Find the sum of first 4 terms of 1, 2, 4, 8, ...
3. Identify the common ratio of 81, 27, 9, ...

Solutions: AP Practice

7^{th} term of 5, 8, 11...

$a_7 = 5 + (7-1) \times 3 = 23$

Sum of first 10 terms of 2, 4, 6...

$S_{10} = 10/2 \times (2\times2 + 9\times2) = 110$

Common difference of 15, 12, 9...

$d = 12 - 15 = -3$

Solutions: GP Practice

5^{th} term of 3, 9, 27...

$a_5 = 3 \times 3^4 = 243$

Sum of 1, 2, 4, 8...

$S_4 = 1 \times (2^4-1) / (2-1) = 15$

Ratio of 81, 27, 9...

$r = 27 \div 81 = 1/3$

Let's Review! Quiz Time

1. In AP 4, 7, 10, ... the common difference is ___

2. The 3rd term of GP 2, 6, 18 is ___

3. AP formula: $a_n = a_1 + (n–1) \times$ ___

Thank You for Learning!